PHY202 Practice Exam on chapter(s): 18, 19

Name:____

These questions assess your understanding of the material from Chapters 18, 19 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

Note that I provide references on practice exams so you can find additional help if you cannot solve a problem; I do not give this information on in-class exams. The related chapter and section are provided after each question. E.g. "(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)" means that the question was inspired by OpenStax College Physics First Edition's Chapter 18, Example 4, which is found within Chapter 18, Section 5. See the end of this practice exam for a list of the sources.

1. When a 3.9-V battery runs a single $126.\text{-W}$ light, how many electrons pass through it each second?
(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ex19_2)

   ANSWER: ____________________________

2. Two identical charged spheres, each with a mass of $25.\ \text{g}$, hang in equilibrium. They are suspended by a string of length $5.7\ \text{cm}$ and are separated by a horizontal distance of $18.\ \text{mm}$. Calculate the magnitude of the charge on each sphere.
(Chapter 18, Section (3) Coulomb's Law - PSE_Serway_4e_ex23_5)

   ANSWER: ____________________________

3. Calculate the magnitude of the force exerted on a 0.41-nC charged particle placed in a $1.8\times10^{4}\text{-}\frac{\text{N}}{\text{C}}$ electric field.
(Chapter 18, Section (4) Electric Field - cp1e_ex18_3)

   ANSWER: ____________________________

4. Calculate the direction of an electric field at the origin due to the presence of two point charges q_1 and q_2. q_1 is located $3.1\ \text{cm}$ north of the origin and q_2 is located $0.14\ \text{cm}$ west of the origin. $q_1 = +4.8\ \text{nC}$ and $q_2 = -4.7\ \text{nC}$. Report your answer in degrees.
(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)

   ANSWER: ____________________________

5. Calculate the final speed of a $4.1\times10^{-18}\text{-C}$ charged particle that accelerates from rest through a potential difference of $2.7\ \text{V}$. The mass of the charged particle is $1.2\times10^{-30}\ \text{kg}$.
(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ex19_3)

   ANSWER: ____________________________

6. Find the total capacitance for three capacitors connected in series, given that their individual capacitances are $4.4\ \text{nF}$, $11.\ \text{nF}$, and $21.\ \text{nF}$.
(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ex19_9)

   ANSWER: ____________________________

7. Suppose a 1.1-V custom battery can move $5.8\times10^{3}\ \text{C}$ of charge and a 12-V car battery can move $5.7\times10^{4}\ \text{C}$ of charge. What is the ratio of work done (or energy delivered) from the car battery compared to the custom battery?
(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ex19_1)

   ANSWER: ____________________________

8. What is the electric potential $1.1\ \text{cm}$ away from the center of a metal sphere that has a 9.7-nC static charge?

(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cp1e_ex19_6)

ANSWER: ____________________________

9. A circuit has two branches: one branch has capacitors 1 and 2 while the second branch has capacitors 3 and 4. Capacitors 1 and 2 are connected in series and capacitors 3 and 4 are connected in series. The two branches are connected in parallel. Draw this circuit and find the total capacitance given that $C_1 = 8.3\,\text{nF}$, $C_2 = 1.0\,\text{nF}$, $C_3 = 2.2\,\text{nF}$, and $C_4 = 4.0\,\text{nF}$. Note: you must draw the circuit correctly to receive credit.
(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ex19_10)

ANSWER: ____________________________

10. Dry air will support electric fields up to its dielectric strength, $3.0 \times 10^{6}\,\frac{\text{V}}{\text{m}}$. Above that value, an electric field creates enough ionization in the air to make the air a conductor, which causes a discharge or spark to reduce the field. Calculate the maximum voltage between a charged region at the base of a thundercloud and the ground if they are separated by $2.1\,\text{km}$ of dry air. Model the cloud and ground as two parallel conducting plates.
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cp1e_ex19_4)

ANSWER: ____________________________

11. Calculate the magnitude of the electrostatic force between an electron and an electron separated by $4.85\,\text{pm}$.
(Chapter 18, Section (3) Coulomb's Law - cp1e_ex18_1)

ANSWER: ____________________________

12. Three charges lie along the x-axis. The positive charge $q_1 = 1.3\,\mu C$ is at $x_1 = 0.96\,\text{m}$ and the positive charge $q_2 = 6.2\,\mu C$ is at the origin $x_2 = 0\,\text{m}$. Where must a negative charge q_3 be placed on the x-axis such that the resultant force on it is zero?
(Chapter 18, Section (3) Coulomb's Law - PSE_Serway_4e_ex23_3)

ANSWER: ____________________________

13. An electron gun has parallel plates separated by $2.0\,\text{cm}$ and gives electrons $43.\,\text{keV}$ of energy. What force would the electric field between the plates have on a piece of plastic that has a charge of $6.7\,\mu C$?
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cp1e_ex19_5)

ANSWER: ____________________________

14. Suppose a tiny drop of oil has a mass of $5.8\,\text{fg}$ and is given a positive charge of $2.1 \times 10^{-17}\,\text{C}$. Calculate the electric force on the drop if there is an upward electric field of strength $1.3 \times 10^{4}\,\frac{\text{N}}{\text{C}}$ due to static electricity.
(Chapter 18, Section (8) Applications of Electrostatics - cp1e_ex18_5)

ANSWER: ____________________________

15. Calculate the strength of an electric field created by a 2.5-nC charged particle at a distance of $4.0\,\text{mm}$ away.
(Chapter 18, Section (4) Electric Field - cp1e_ex18_2)

ANSWER: ____________________________

16. Suppose $62.\,\text{mm}$ separates two 1.4-m❑^{2} plates in a parallel-plate capacitor. What charge is stored in this capacitor if it is connected to a 1.1-V battery?
(Chapter 19, Section (5) Capacitors and Dielectrics - cp1e_ex19_8)

ANSWER: ____________________________

17. A heart defibrillator delivers 3.9×10^{3} J of energy by discharging a capacitor initially at 1.9×10^{4} V. What is its capacitance?
(Chapter 19, Section (7) Energy Stored in Capacitors - cp1e_ex19_11)

ANSWER: ____________________________

18. A particular Van de Graaff generator has a metal sphere of diameter $72.\mathrm{cm}$ that produces a voltage of $176.\mathrm{kV}$ near its surface. What excess charge resides on the sphere?
(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cp1e_ex19_7)

ANSWER: ____________________________

cp1e: College Physics (1st Edition) by OpenStax
PSE_Serway_4e: Physics for Scientists and Engineers (4th Edition) by Serway
tkd: I did not refer to anything while writing this question